

Advanced Machine Learning

DATA 642

Lecture 15 — Evaluating GANs

*Figures and notes are from <https://towardsdatascience.com/on-the-evaluation-of-generative-adversarial-networks-b056ddcdfd3a> and the paper <https://arxiv.org/abs/1802.03446>

Evaluation of GANs

Evaluation of Generative Adversarial Networks (GANs) presents challenges due to the absence of an objective loss function for training the generator. As a result, assessing the training progress and the quality of the model solely through loss metrics is not feasible. Instead, a range of qualitative and quantitative methods have been developed to gauge the performance of GAN models, primarily focusing on the quality and diversity of the synthetic images generated.

Qualitative Evaluation

Since GANs lack an objective loss function, their evaluation relies on assessing the quality of the synthetic images produced. A fundamental approach involves manually inspecting and judging the generated examples at various stages of training.

However, this method presents several limitations:

- **Subjectivity:** Evaluation is influenced by the biases of the reviewer.
- **Domain expertise:** Distinguishing realistic from unrealistic images requires specialized knowledge. For instance, in the context of dermatology, input from dermatologists is crucial for assessing synthetic examples accurately.
- **Limited scalability:** The number of images that can be reviewed manually is restricted.
- **Lack of standardized procedures:** There is no universally accepted method for qualitatively evaluating generated images, as it often varies depending on the specific application.

Quantitative Evaluation

In this section, we will explore metrics commonly employed to evaluate generative models during training. These metrics complement qualitative assessments, offering a comprehensive evaluation of GAN models.

When assessing the performance of our GAN, we primarily consider several key properties:

- **Fidelity:** This refers to the quality of the generated samples, indicating how realistic they appear. Fidelity metrics measure the dissimilarity between each fake sample and its nearest real counterpart.
- **Diversity:** This relates to the variety of generated samples, indicating how well the generated images cover the entire diversity of the real distribution.
- **Authenticity/Generalizability:** This measures the rate at which the model invents new samples and tries to spot overfitting to the real data.
- **Predictive Performance:** This evaluates how well the synthetic data covers the distribution of the real data. We can also train and test models with synthetic data, allowing us to assess whether synthetic data preserves the ranking of predictive models trained on real data.

Fréchet Inception Distance (FID)

- **Overview:**

- FID compares real and fake images by using classifiers as feature extractors.
- The Inception-v3 classifier, pretrained on ImageNet, is commonly used.
- Removing the output layer yields embeddings for real and fake images.
- These embeddings represent two multivariate normal distributions, compared using the Multivariate Normal Fréchet Distance or Wasserstein-2 distance.

- **Mathematical Details:**

- The Multivariate Normal Fréchet Distance is calculated as the squared Euclidean distance between the means of the two distributions, plus the trace of the sum of their covariance matrices minus twice the square root of the product of their covariance matrices.

- **Shortcomings:**

- Relies on a pre-trained Inception model, which may not capture all features accurately.
- Requires a large sample size for accurate evaluation.
- Slow to run and utilizes limited statistics, focusing only on mean and covariance.

Kernel Inception Distance (KID)

- Proposed as an alternative to FID (Fréchet Inception Distance).
- Addresses the issue of biased estimation in FID, especially prominent in smaller datasets.
- Unlike FID, KID's expected value remains consistent regardless of the dataset size, making it more suitable for smaller datasets.
- It offers computational efficiency, numerical stability, and simplicity in implementation compared to FID.

Inception Score

- **Overview:**

- Measures both diversity and fidelity of generated images, similar to FID and KID.
- Utilizes a pre-trained Inception v3 model to predict class probabilities for each generated image.
- A higher score indicates more realistic and diverse generated images.

- **Mathematical Details:**

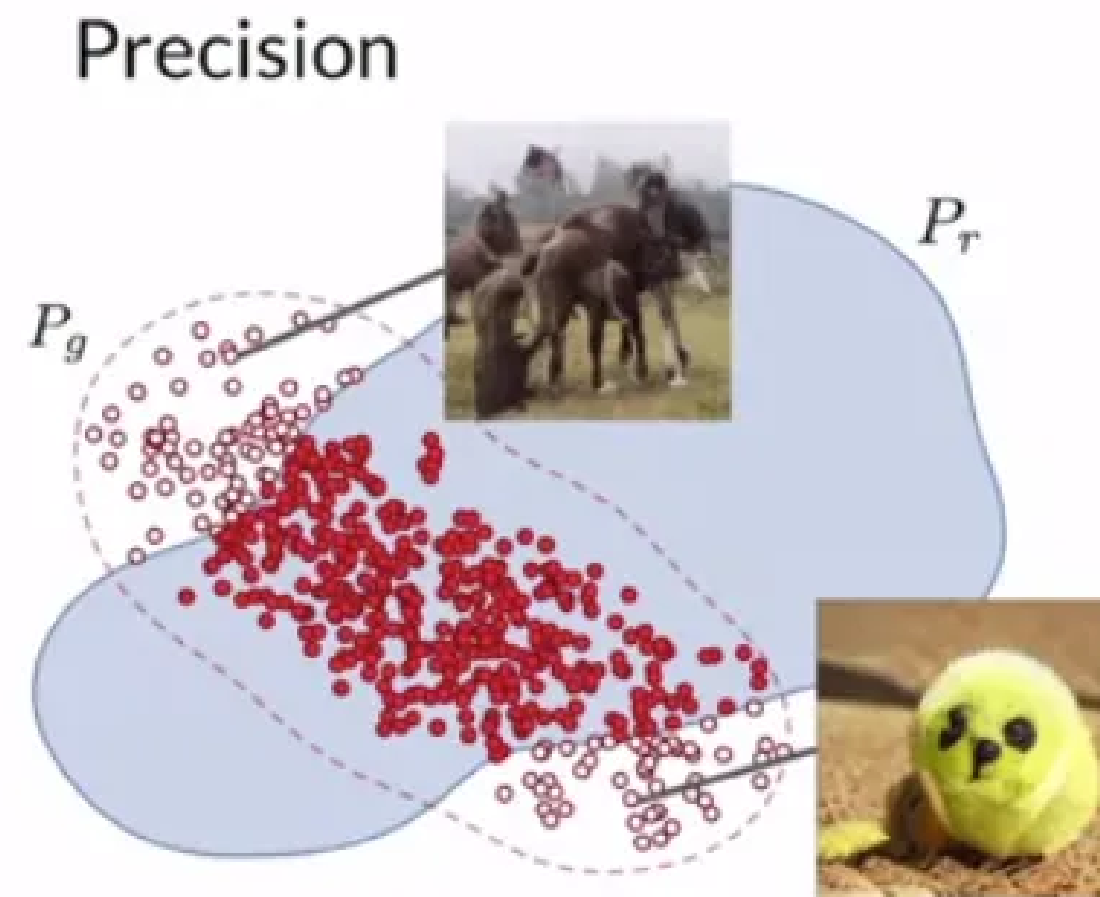
- The Inception Score combines two components: conditional probabilities and marginal probabilities.
- Conditional probabilities measure the confidence of the classifier in assigning a specific class label to each generated image.
- Marginal probabilities assess the variety across generated images, with higher entropy indicating more diversity.
- These components are combined using the Kullback-Leibler divergence (KL divergence), quantifying the difference between the conditional and marginal probability distributions.

- **Limitations:**
 - Only evaluates fake images, without comparison to real ones.
 - Performance constrained by Inception classifier capabilities and training data.
 - Attempts to capture fidelity and diversity but may not be as effective as FID in some cases.

Precision

Precision focuses on fidelity, measuring the overlap between real and fake data relative to non-realistic images generated by the generator.

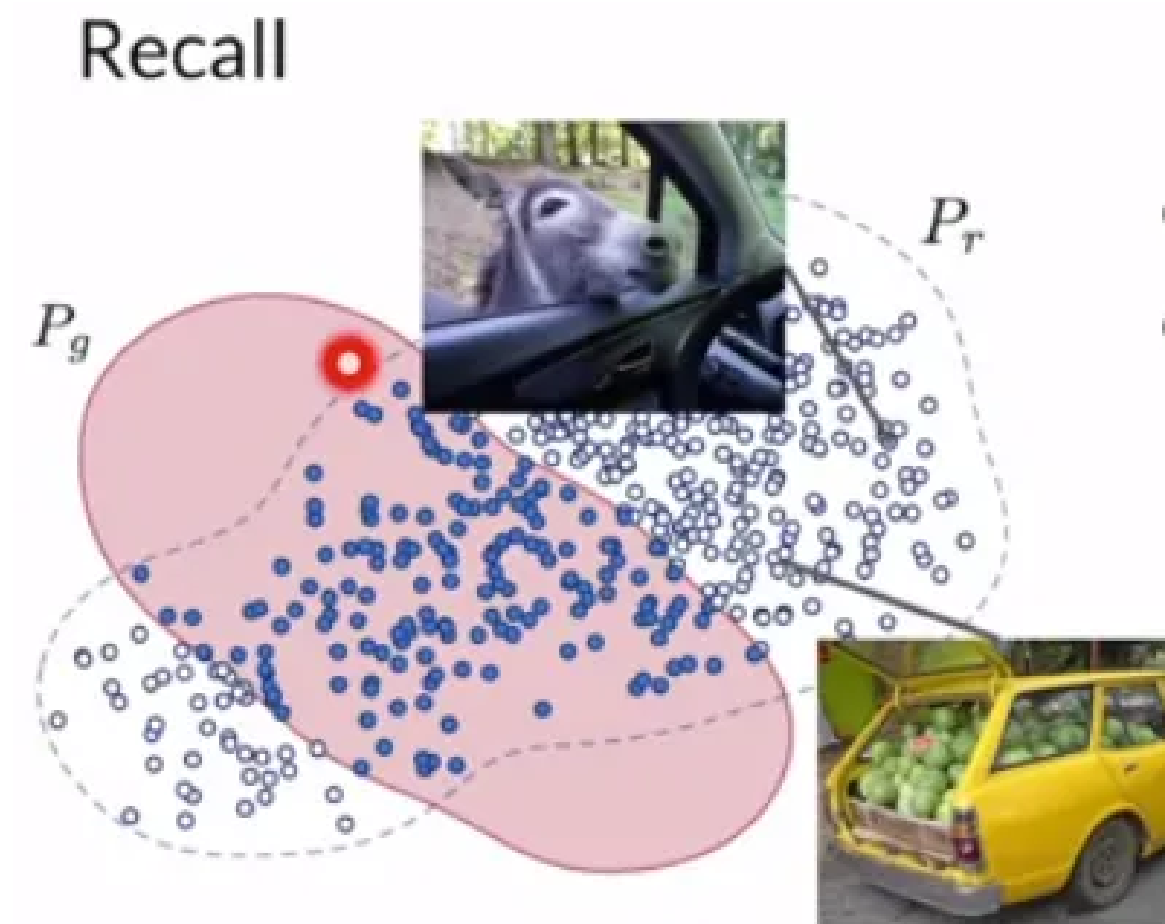
- Evaluates the fidelity of generated images.
- Measures the overlap between real and fake data compared to non-realistic images generated.
- High precision indicates minimal generation of non-realistic images.



Recall

Recall assesses diversity, measuring the overlap between real and fake data relative to all the real data that the generator fails to model.

- Assesses the diversity of generated images.
- Measures the overlap between real and fake data compared to all the real data that the generator fails to model.
- High recall indicates better coverage of the real data distribution.



Conclusion

- **Evaluation Metrics for GANs:**
 - The metrics discussed in this lecture are among the commonly used ones in the field.
 - It's important to note that there are many other metrics available for evaluating GANs, depending on specific research goals and applications.
- **Further Exploration:**
 - Researchers are continually developing new metrics and refining existing ones to improve the evaluation process for GANs.
 - Exploring additional evaluation metrics can provide deeper insights into the performance and behavior of GAN models.